Robot Navigation Experiment Report

Experiment: Live three-model baseline vs optimized benchmark

Status: completed

Application version: 0.1.0

Executive summary

The experiment recorded 24 runs across 6 model/configuration groups. 12 runs met both interpretation and route-validity criteria. Detailed rows, including failures, remain available in JSON and CSV exports.

Observed findings

Within this experiment, baseline groups produced 0 verified successes across 12 runs, while optimized groups produced 12 across 12 runs. The lowest observed optimized group mean was 5710.82 ms for qwen3:0.6b. Optimized groups recorded 6 cache hits in total. These observations apply only to the recorded hardware, scenarios, and sample size.

Model comparison

Model	Mode	Runs	Success	Mean latency	Failures
qwen2.5:0.5b	baseline	4	0.0%	24162.42 ms	4
qwen2.5:0.5b	optimized	4	100.0%	7228.01 ms	0
qwen3:0.6b	baseline	4	0.0%	21694.08 ms	4
qwen3:0.6b	optimized	4	100.0%	5710.82 ms	0
smollm2:1.7b-instruct-q4_K_M	baseline	4	0.0%	38042.98 ms	4
smollm2:1.7b-instruct-q4_K_M	optimized	4	100.0%	14738.41 ms	0

Methodology

Success requires a correct structured interpretation and a validator-approved A* route. Failures remain in denominators. Offline tests are not model results.

Limitations

Results depend on the tested hardware, exact model tag, quantization, warm-up state, scenarios, and sample size.

Reproducibility

Experiment ID: ad5f9bb6-bb44-43af-8926-847361f5ab62. Inspect JSON/CSV exports and versioned scenarios for raw evidence.